

Grade 6 Accelerated  
Unit 14  
Lesson 8 Practice

Name Answer Key  
Date \_\_\_\_\_  
Period \_\_\_\_\_

Write and solve an algebraic equation for each of the following.

1. At a restaurant, 9 friends decided to split the bill evenly. If the total bill was \$170.28, how much would each friend pay?

$$9x = 170.28 \quad x = \$18.92 \text{ each}$$

2. A bride-to-be is stuffing wedding favor bags using a random mix of her 500 pieces of party favors to split evenly among her guests that attend the wedding. If each guest received 20 pieces, how many guests attended the wedding?

$$\frac{500}{g} = 20 \quad g = 25 \text{ guests}$$

3. Jake has a budget of \$200 for groceries each month. How much is Jake's budget for groceries for an entire year?

$$\frac{x}{12} = 200 \quad x = \$2400 \text{ for the year}$$

4. During a week, Brodie earned \$58.40 for walking dogs. If Brodie walked 8 dogs, how much did he charge for each walk?

$$8x = 58.40 \quad x = \$7.30 \text{ for each walk}$$

5. At a local arcade, the number of tickets needed for a stuffed bear is 20,000 tickets. If Darnell earns 500 tickets each time he visits the arcade, how many visits will it take to get the stuffed bear?

$$500t = 20,000 \quad t = 40 \text{ visits}$$

6. Deborah has 80 books that she wants to put on her new 5-shelf bookshelf. If she distributes the books evenly to each shelf, how many books will be on each shelf?

$$5b = 80 \quad b = 16 \text{ books to each shelf}$$

7. A recipe for pancakes calls for 2 cups of water for each batch. If someone wants to make 3 batches, how many cups of water will be needed?

$$3(2) = c \quad c = 6 \text{ cups of water}$$

8. A car manufacturer offers 9 different car models, each with 6 color options. How many car choices would a customer buying from that manufacturer have?

$$9(6) = c \quad c = 54 \text{ car choices}$$

9. A shoe storage bin can hold 8 pairs of shoes. Kiera purchased 4 bins, how many pairs of shoes can she store?

$$x = 8(4) \quad x = 32 \text{ pairs}$$

10. Minnie is selling lemonade to raise money. At the end of the day, she raised \$10.60. If each cup sold for \$0.25, how many cups of lemonade did Minnie sell?

$$0.25c = 10.60 \quad c = 42.4 \text{ cups or } c \approx 42 \text{ cups}$$